TheONE AI

Voice-First AI Companion • Natural & Savage Comedy Architecture

Live App: theone-ai.onrender.com
GitHub: github.com/Tarak098/TheONE-AI
Platform: Render Web Service (FastAPI Full-Stack)

1. EXECUTIVE OVERVIEW

TheONE AI is a minimal, voice-first companion inspired by ChatGPT's Advanced Voice Mode. It centers verbal interaction around an audio-reactive **Central Voice Orb**. Powered by a **Natural & Savage Comedy Persona**, it talks like that one hilarious friend in real life: simple everyday spoken English, zero complicated vocabulary, instant comebacks, and a strict **one-sentence limit** (≤ 15 words).

2. END-TO-END VOICE PIPELINE

```
[ User Microphone ] ──► (16 kHz Linear PCM Mono WAV)
        │
        ▼
[ Gnani Prisma STT v3 ] ──► Fast BCP-47 Speech Recognition
        │
        ▼
[ Intelligence Orchestrator ]
 ├── Natural Savage Comedy Persona (Simple Spoken English)
 ├── Context-Aware Roaster (Procrastination, exams, money, ego)
 ├── Google Gemma 4 26B (Gemini 2.0/1.5 Flash Resilient Failover)
 └── Strict Response Validator (≤ 1 Sentence, 3–10 Words)
        │
        ▼
[ Gnani Timbre TTS v2.5 ]
 ├── Voice: "Yashvi" (Female, Cheerful 1.04x speed prosody)
 └── Voice: "Zayan" (Male, Deadpan 0.98x speed prosody)
        │  (48 kHz Mono WAV Audio)
        ▼
[ Web Audio API Player ] ──► Real-Time AnalyserNode ──► Voice Orb
```

3. NATURAL HUMAN HUMOUR & SAVAGE ROASTS

TheONE AI abandons robotic tropes and complicated words. It speaks plain, conversational English with spontaneous comebacks:

Daily Conversational Examples:
- *Study tomorrow:* "Bro, tomorrow has blocked you already."
- *Woke up at 12:* "Good morning, grandpa."
- *No money:* "Your bank account is running a charity for zeroes."
- *Very intelligent:* "Yeah bro, your mirror must be proud."
- *Scrolling all day:* "Your thumb deserves employee of the month."
- *No replies:* "Even your notifications gave up."
- *Gym promise:* "Your gym membership is already crying."
- *Failed exam:* "Bro, the question paper won this fight."

4. CORE CAPABILITIES

- **Central Voice Orb:** 5-state Web Audio visualizer (`IDLE`, `LISTENING`, `THINKING`, `SPEAKING`, `ERROR`) reacting to microphone and speech waveform volume.
- **Simple Everyday English:** Common daily words only. Zero robotic jargon, formal vocabulary, or textbook jokes.
- **Independent Voice Selection:** 1-click toggle between **Yashvi** (Female) and **Zayan** (Male) in top navigation bar, saved to `localStorage`.
- **Strict 1-Sentence Constraint:** Maximum 1 sentence (≤ 15 words). No paragraphs, no explanations.
- **Seamless Scrolling UI:** Global side scrollbar guarantees full accessibility to the chat field on all screen sizes.

5. PRODUCTION TECH STACK

Component	Technology
Backend	FastAPI 0.110+, Python 3.11, Uvicorn ASGI
Frontend	HTML5, CSS3, Vanilla JS (ES6+), Web Audio API
STT Engine	Gnani Prisma v3 REST API (vachana.ai)
LLM Engine	Google Gemma 4 26B (Google GenAI SDK)
TTS Engine	Gnani Timbre v2.5 REST API (48 kHz WAV)
Deployment	Render Web Service (Full-Stack ASGI on HTTPS)

6. RENDER DEPLOYMENT SPECS

- **Architecture:** Single ASGI service mounts `/frontend` statically, eliminating CORS and proxy latency.
- **Build Command:** `pip install -r requirements.txt`
- **Start Command:** `uvicorn main:app --host 0.0.0.0 --port $PORT`
- **Health Check:** `GET /health` returns `{"status":"healthy"}`
- **Microphone Access:** Browser Web Audio & MediaRecorder work natively over Render's HTTPS origin.